

Costing How an NPC Reads: Attributability as a Proxy for Believable Search Behaviour

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Independent

<https://github.com/SL-A-SH/AI-Native-Cohort>

Abstract

A non-player character searching for a hidden player must act on a belief rather than on knowledge. Occupancy-map methods for this are established in game AI. We take that machinery as given and ask a narrower question: can the way behaviour reads to an observer be priced as a cost, alongside the tactical costs an agent already pays? We define three such terms (illegibility, redundancy, dither) and an attributability measure computed by running a second belief that receives no positive observations and comparing the two. On a single hand-made grid, across 8 case-generation seeds and 40 scenarios each, the resulting agent is beaten on capture rate by two much cheaper scripted policies in every seed, while re-searching already-cleared ground roughly a tenth as often. A factorial ablation shows that the negative-information update and a hand-coded anti-redundancy memory are substitutes rather than complements: either alone produces most of the effect, and the hand-coded version produces the best redundancy figure and the worst capture rate. We report these results as an engineering study. We do not claim that attributability measures believability: no human has judged this agent, and validating the measure against observer judgement is the experiment the work most needs.

1 Introduction

An enemy that always knows where you are is not difficult, it is unpleasant.¹ An enemy that never finds you is not fair, it is broken. Between those two failures sits the behaviour designers actually want, and it is usually produced by hand.

¹This is a course preprint, prepared as coursework and formatted with the IJCAI-ECAI-26 author kit. It has not been submitted to IJCAI or to any other venue, and no claim of review or acceptance is made.

This paper concerns one narrow part of that problem. Given an NPC that maintains a probability distribution over a hidden player’s location and chooses actions by minimising expected cost, we ask whether the legibility of its choices can be priced in the same objective as its tactical costs, rather than being left to animation and scripting.

A retracted motivation. This work began from a claim that turned out to be false, and we state it plainly because the correction shapes everything that follows. We believed that shipped game AI represented belief only discretely, as a remembered position plus an awareness level, and that probabilistic occupancy methods lived in robotics. That is wrong. Isla introduced occupancy maps to a game AI audience two decades ago [Isla, 2006] and later shipped a commercial stealth game built around them [Isla, 2013]. The grid, the diffusion over walkable space, and cell-scope negative information are prior art in games, not a transfer from robotics.

What remains is narrower. The belief machinery here is a replication. The contribution, such as it is, is the cost layer over it and the attributability measure, both evaluated honestly and both reported with negative results where we found them.

Contributions.

1. A cost formulation with three terms that price how behaviour reads rather than how accurate it is, each traceable to a specific practitioner complaint (Section 3).
2. An attributability measure: the ratio between the agent’s belief and a counterfactual belief deprived of positive observations (Section 4).
3. An evaluation against three controls in which the proposed agent loses on the primary task, and a factorial ablation showing that its clearest advantage is not caused by the mechanism we assumed (Sections 6 and 7).

2 Related Work and User Discussions

Belief in shipped game AI. Isla’s Halo 2 work describes a discrete model: a remembered target position and orientation plus awareness levels, with the actor explic-

itly permitted to believe things that are not true [Isla, 2005]. F.E.A.R.’s published AI work is goal-oriented action planning [Orkin, 2006]. The Alien: Isolation director holds the player’s true position and deliberately tells the alien only a general area [Thompson, 2017]. Against these, Isla’s occupancy-map work [Isla, 2006; Isla, 2013] is the direct antecedent of the belief model used here. The underlying filter is standard recursive state estimation [Thrun et al., 2005].

User discussions. Three of the design decisions in this paper came from public discussion with practitioners and players rather than from literature, and we record which.

A roguelike developer described a shipped give-up rule in his own game: a monster follows the player into a room, and “if that room has two doors and I exit, it will give up after it sees the room empty.” This retracted our assumption that negative information is generally absent from game AI. The distinction that survives is narrower and is what we implement: their rule is room-scope and terminates a pursuit, ours is cell-scope and redistributes probability.

A player observed that an omniscient enemy is acceptable when the fiction explains it — a mind-reader may know where you are — while an ordinary guard doing the same thing is infuriating. Identical information, opposite reaction. This reframed our objective: the complaint is not about how much the NPC knows but about whether an observer can attribute the knowledge to something. The attributability measure is our attempt to make that computable.

A second developer noted that his monsters enter a “where did they go?” state in which they look around and pace back and forth to show confusion, before returning to patrol. We had recorded that oscillation as our worst failure mode; he animates it deliberately. The failure is therefore not oscillation but oscillation that is unbounded and unexplained.

3 Agent Design

Problem. The agent observes a stream of noisy perception events (a sighting, a sound, a door state change) and the time elapsed since each. It must choose an action because the player’s position is not known.

Input and hidden state. The hidden state is the player’s cell on a 24×14 grid with 201 walkable cells. The agent never receives the player’s position. Its inputs are observations and its own location.

Actions. Six: `search` a cell, `hold_position`, `flank` to a chokepoint, `call_reinforcements`, `retreat`, and `resume_post`, the only terminal action. Two of these are never chosen in our experiments and we report that as a finding rather than trimming the action set (Section 8).

Cost terms. Each option’s cost is a sum. Three terms are tactical: travel, a per-tick cost of the player remaining at large, and an expected gain crediting belief mass brought under observation. Three price how the behaviour reads:

Illegibility penalises acting where the agent’s own evidence does not point, scaled by the unattributable fraction defined in Section 4.

Redundancy penalises returning to ground swept within the last 25 ticks, decaying with age. This is a behavioural memory, deliberately separate from the belief.

Dither penalises abandoning a target before arriving at it.

Human reasoning function. The agent identifies its own uncertainty: holding and standing down are priced like every other option, so “I do not know enough to commit” emerges from the comparison rather than from a tuned entropy threshold.

A caution on the weights. The three believability weights are 6, 8 and 4. They are assumed, not fitted. They are also not independently meaningful: a constant lookahead multiplier scales the gain terms but not travel, dither or redundancy, so only ratios among the coefficients affect decisions. We report sensitivity rather than defending the values.

4 Probability Model and Decision Rule

The belief b_t over cells alternates predict and update. Prediction diffuses mass to walkable neighbours, so probability flows around walls rather than through them:

$$b_t^-(x) = \alpha b_{t-1}(x) + (1-\alpha) \sum_{x' \in N(x)} \frac{w(x)}{\sum_{y \in N(x')} w(y)} b_{t-1}(x'), \quad (1)$$

with $\alpha = 0.55$ the probability of not moving and w a transition weight (uniform in policy A; raised on chokepoints in policy B). Update is Bayes’ rule with an unnormalised likelihood:

$$b_t(x) = \frac{L(z_t | x) b_t^-(x)}{\sum_{x'} L(z_t | x') b_t^-(x')}. \quad (2)$$

A sighting uses a tight Gaussian about the reported cell, a sound a wider one, both with a false-positive floor, and distance is measured through the map rather than across it. A failed search multiplies each searched cell by a miss rate of 0.15, so looking and seeing nothing reduces a cell without eliminating it.

Attributability. Alongside b_t we run b_t^0 , the same filter receiving diffusion and failed searches but no positive observations. Their ratio

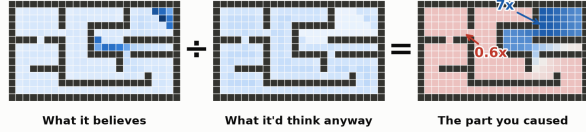
$$r_t(x) = b_t(x) / b_t^0(x) \quad (3)$$

is 1 where the agent would have believed as much anyway. The illegibility term is priced off $1/r$, capped at 3.

What this quantity is not. r is a counterfactual belief ratio. It is not a decomposition of posterior mass into player-caused and non-player-caused parts, and a positive observation can move it through the normalising constant alone. We previously described $1 - 1/r$ as an “attributable fraction”; that description is withdrawn.

How to catch an AI guessing

Run the guard's map of you twice. Once with ears and eyes, once deaf and blind.



Above 1, you put it there.

Below 1, it is guessing.

After this guard searched the room and found nothing, 157 of the 201 floor tiles fell below 1. Almost anywhere it goes next, it is making up.

Day 5 of 7. Every tile and both numbers are real output from the filter.

Figure 1: The attributability construction, on one logged state after a failed search. Left: the agent’s belief. Centre: the same filter with every sighting and sound withheld, leaving only diffusion and failed searches. Right: their ratio on a log scale. Blue marks cells where positive observations raised the belief above the counterfactual; red marks cells where the evidence pointed elsewhere and the agent believes less than it would have anyway. In this state 157 of 201 walkable cells fall below 1.

Decision rule. The policy is $\arg \min$ over expected cost. There is no threshold on belief. The scalar being minimised is a designer utility, not an objectively derived cost: it sums failure-ticks, walking steps and three invented weights, and nothing in the probability model licenses those units as commensurable.

5 Test Method

Forty scenarios, spread deliberately across five guard posts and four bands of initial guard-to-player distance rather than sampled at random. Each episode opens with one noise from the player’s starting cell and runs to capture, stand-down, or 80 ticks. The player walks to randomly chosen goals with a 15% random-step probability and does not react to the agent. Every scenario is replayed identically for every arm, making the comparison paired. Sweeps repeat all of this across 8 case-generation seeds.

The label is hidden at decision time. The player’s true position is used for exactly three things: emitting observations, supplying the degraded oracle, and scoring afterwards. No argument to the agent’s decision function carries it.

Controls. last-known-position records the last observed cell and goes there. scripted-sweep does that, then sweeps nearby chokepoints on a timer. degraded oracle receives the player’s true cell, gated by line of sight, a reaction delay and a position error. The third is a control, not a competing method: its information set is different in kind.

Measurement. An external shadow filter, driven only by the observations the player caused, scores every arm’s

Arm	Capture	Redundant	Attrib.
Belief (policy A)	72.2 $\pm$ 7.8	10.9 $\pm$ 5.4	13.9
Belief (policy B)	71.2 $\pm$ 5.6	9.8 $\pm$ 6.9	13.2
Last-known-position	79.1 $\pm$ 5.1	114.5 $\pm$ 27.7	22.2
Scripted sweep	83.7 $\pm$ 6.8	136.9 $\pm$ 37.9	21.9
Degraded oracle	62.8 $\pm$ 5.4	248.9 $\pm$ 35.5	11.4

Table 1: Eight case-generation seeds, 40 scenarios each; mean $\pm$ SD across seeds. Capture is a percentage; redundant searches are counts per 40 episodes.

Neg. info	Memory	Redundant/commit	Capture
off	off	0.363 $\pm$ 0.021	73.4
on	off	0.029 $\pm$ 0.010	71.9
off	on	0.002 $\pm$ 0.002	65.9
on	on	0.014 $\pm$ 0.007	72.2

Table 2: Factorial ablation, 8 seeds $\times$ 40 scenarios per cell. Neither switch requires a code change: negative information is disabled by setting the miss rate to 1, the memory by setting its weight to 0.

attributability on the same yardstick, including the three that hold no distribution of their own.

6 Results

Table 1 is the main result and it is not favourable. The belief agent is beaten on capture by both scripted controls in every seed, by 6.9 and 11.6 points on average. It re-searches cleared ground roughly a tenth as often as any control. We treat “0 of 8 seeds” as descriptive: eight seed-level aggregates are too few to support a claim about generalisation, and we have not specified an inferential model that would license one.

The redundancy advantage is not caused by what we assumed. The agent contains two mechanisms that could reduce re-searching: the probabilistic negative-information update, and the hand-coded redundancy penalty. A factorial ablation (Table 2) separates them.

Three findings. Both mechanisms work: removing both raises redundancy 26-fold. Negative information works alone, cutting redundancy from 0.363 to 0.029 with the memory penalty switched off entirely. But they are substitutes, not complements — either alone achieves most of the effect and together is no better than either. And the hand-coded memory alone produces the best redundancy figure in the table and the worst capture rate: reading that column by itself would select the worst agent. Both believability mechanisms cost capture; the arm with neither has the highest capture rate we measured.

Calibration. When the agent assigns above 20% to its peak cell, the player is in that exact cell 20.3% of the time; at a stated 14.4%, 11.6%. The model is overconfident by roughly 1.7 $\times$ at the top of its range. Because gains are linear in belief mass while other costs are fixed, rescaling the belief changes the $\arg \min$: miscalibration

therefore does bear on the policy, and an earlier claim of ours that beliefs here are “ordinal, not cardinal” is withdrawn.

7 Failure Analysis

We name six failure conditions and examine instances of each; all are logged by scenario and tick in the repository.

F1, watching-not-catching (136 cases). The agent holds while the player is within reach. In one logged instance the guard stands two cells from the player: holding costs 1.00 while every nearby search costs twelve, because the redundancy term charges for ground just swept. We initially diagnosed this as a design tension. A reviewer identified a genuine bug underneath it — gain excluded recently-looked-at cells while the filter had already reduced them, applying negative information twice. Fixing it improved capture, precision, calibration and escalation, and did not fix F1, which rose from 107 to 136. The candidate causes we cannot yet separate are: capture is not the optimised event, so being in reach has no value under the objective; the action set may lack a “close the last two cells” move; and capture is evaluated after the player moves while findability is scored before.

F2, chasing-the-smear (271 cases) is the most common and the most forgivable: mean attributability 4.65, so these commits were evidence-backed and read as honest mistakes. F3, illegible-commit (133) is the one that reads as cheating by our own proxy. F5, premature stand-down occurs once in 200 episodes. F4 and F6 no longer occur.

Which error costs most. In outcome terms, false positives: 6.8 per episode against 3.4 false negatives. In believability terms we argue the reverse, because F2’s errors are attributable and F1’s are not. That ordering is an argument, not a measurement, and it is exactly what a human study would settle.

Ten bugs. The repository records ten defects found during development, four of which changed a headline number; two were found by AI review rather than by us. We report this because it bears on interpretation: behaviour here is demonstrably sensitive to bookkeeping choices, which weakens any reading of these results as a stable scientific finding.

8 Limitations, Ethics and Human Control

No human has judged this agent. The dependent variable the paper cares about is how behaviour reads to an observer, and we measure a proxy instead. This is the central validity limitation and it is not cosmetic.

One map. Chokepoints, points of interest, distances and visibility are all derived from a single hand-made grid whose corridors are one cell wide. Several results may be properties of that topology.

A non-reactive player. The player never evades, never doubles back and never manufactures a decoy. The controls that win here do so by walking to the freshest observation, which is the strategy an evading player punishes

most directly. This is the experiment we would run next, and it is not merely an extension: it is directly relevant to whether attributability matters at all.

Roughly fifteen assumed parameters, none fitted. Not fitting them means we did not optimise against our own results; it does not make them justified.

A biased primary metric. Scoring a decision by whether the player is findable at the chosen target at that instant favours policies that chase the newest observation. It also penalises flank for doing what its own definition specifies. Task-aligned temporal measures should replace it.

Two dead actions. retreat is strictly dominated because nothing in the simulation can harm the agent, and flank is never chosen. A third, call reinforcements, is priced with a benefit the simulation cannot deliver, since no second agent exists.

Two indistinguishable policies. Policy B changes the belief without changing behaviour, and the player generator shares policy B’s own chokepoint bias, so it is not an independent test.

Ethics and human control. We see no substantive AI-ethics concern here beyond ordinary game design, and we decline to pad one. The relevant issue is player-facing information asymmetry: an NPC that acts on knowledge a player cannot account for damages the informational contract a stealth game depends on, which is the design intuition this work tried and failed to measure. All behaviour is confined to a simulator; nothing is deployed.

9 Conclusion and New Questions

We priced three readings of NPC behaviour as costs and measured what that bought. It bought substantially less re-searching of cleared ground, at a consistent cost in capture rate against two much cheaper scripted policies, and an ablation shows the mechanism responsible is not the one we assumed. We report this as an engineering study.

Three questions we cannot answer:

1. Does attributability predict human judgement? Show observers matched trajectories that differ in evidence-to-action relationship and ask which reads as fair, intelligent, or cheating. Until this is done the measure is a proposal.
2. Does any of this beat an equally cheap scripted controller on observable traces? Our scripted controls already win the primary task. If observers cannot distinguish the two, that is an important negative result.
3. Does it survive a player who evades and plants decoys? Our decision record shows one manually constructed false observation steering the agent confidently to the wrong side of the map. That was a diagnostic, not evidence from the benchmark, and a reactive player is where it would become one.

Contribution Statement

A.R. is the sole author and carried out the problem formulation, implementation, experiments, analysis and writing.

AI Use Statement

Large language models were used throughout: to draft and refactor code, to propose the experimental design, to draft this manuscript, and as reviewers. Three structured AI reviews (practitioner, probability, preprint) were commissioned before writing, and their findings changed the paper substantially: they identified the double-counted negative information behind F1, the single-seed sweep, the confound requiring the ablation in Table 2, the withdrawn “ordinal, not cardinal” claim, and the prior art that retracted the paper’s original motivation. AI also introduced errors that were caught by checking: an early claim that F.E.A.R. and Halo shipped probabilistic belief was false, and the replacement claim in Section 1 was itself false and is retracted here. All results were produced by code in the linked repository; the author verified each number reported.

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